

- 7(a) Longitudinal wave – oscillations of wave particles parallel to direction of transfer of energy of wave [1]
Transverse wave – oscillation of wave particles perpendicular to direction of transfer of energy of waves [1]
- (b) S-waves cannot pass through the liquid outer core. [1]

- (c)(i) Akita – 30 mm
 Pusan – 56 s, 540-550 km,
 Tokyo – 425-435 km, 210 mm

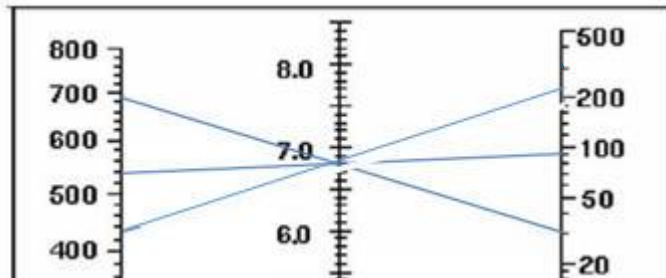
All amplitudes correct [1]

Time interval for Pusan correct [1]

Distance for Pusan (allow ecf) and Tokyo read correctly [1]

- (ii) Understand must draw 3 circles center at stations [1]
 All circles drawn with compass and correct scale [1]
 Epicentre at KOBE [1]

- (iii)1. -Distance from epicenter
 -Scattering at boundary of different materials, cracks etc.
 -Absorption due to rocks in Earth
 Any other suitable answers [1]



Magnitude = 6.8 [2]

- (d)(i) Ratio = $10^9/10^{6.8} = 158$ [1]

1. The distance between the epicentre and the station. [1]
2. The above shows that a change in magnitude of 2.2 correspond to a change in intensity of about 160 times. Hence using a log scale allows us to compress the scale to more manageable numbers [1]

- (ii) Ratio = $10^{(1.5)(2.2)} = 1995$ (2000) [1]

- (e) - Population density of the affected area might be low.
 - The affected area might have stricter building codes that require buildings to have earthquake-proof features.